Exp No:17 Experiment to understand Linear Regression for a given data set.

Description: Understand the Linear regression for the dataset given.

Code:

```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/Salary_Data.csv')
df.head()
```

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891

```
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #  Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0  YearsExperience  30 non-null     float64
 1  Salary          30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
df.dropna(inplace=True)
df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #  Column          Non-Null Count  Dtype
---  ------          --------------  -----
```

0 YearsExperience 30 non-null float64

1 Salary 30 non-null int64

dtypes: float64(1), int64(1)

memory usage: 612.0 bytes

df.describe()

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.556667	78345.133333
std	3.019669	28712.828489
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	5.000000	66983.500000
75%	8.125000	104512.000000
max	11.200000	127345.000000

```
features=df.iloc[:,[0]].values

label=df.iloc[:,[1]].values

from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(

  features, label, test_size=0.2, random_state=42

)

from sklearn.linear_model import LinearRegression

model=LinearRegression()

model.fit(x_train,y_train)

model.score(x_train,y_train)

0.9622248020359685

model.score(x_test,y_test)

0.958069463205837

model.coef_

array([[9343.7944329]])

model.intercept_
```

array([26528.39510872])

import pickle

pickle.dump(model,open('SalaryPred.model','wb'))

model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))

yr_of_exp_NP=np.array([[yr_of_exp]])

Salary=model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {}: ".format(yr_of_exp, Salary))

Enter Years of Experience: 44

Estimated Salary for 44.0 years of experience is [[437655.35015639]]:

Result:

Understood the Linear regression for the dataset given.